

L2.6a Optimization

§3.7 — sort the question, then run the template

Optimization under a constraint

Both of these stop at the sort. Write all three slots — **OEq**, **Constraint**, **Interval** — and if a slot has nothing in it, write **NONE** rather than leaving it blank. Do not solve either one.

1. Two numbers. Find two numbers whose difference is 100 and whose product is a minimum.

2. Photosynthesis. The rate (in mg carbon/m³/h) at which photosynthesis takes place for a species of phytoplankton is modelled by

$$P = \frac{100I}{I^2 + I + 4}$$

where I is the light intensity, measured in thousands of foot-candles. For what light intensity is P a maximum?

The optimization template

Run the whole template on each of the next three. Draw the picture in a box, sort the question, then finish — and answer the question that was asked, in the units it was asked in.

- 3. The farmer and the neighbour.** A farmer wants to fence in a rectangular plot of land adjacent to the north wall of his barn. No fencing is needed along the barn, and the fencing along the west side of the plot is shared with a neighbour who will split the cost of that portion of the fence. The fencing costs \$20 per linear foot to install, and the farmer is not willing to spend more than \$5000. Find the dimensions of the plot that would enclose the most area.

4. **Crop yield.** A model used for the yield Y of an agricultural crop as a function of the nitrogen level N in the soil (measured in appropriate units) is

$$Y = \frac{kN}{1 + N^2}$$

where k is a positive constant. What nitrogen level gives the best yield?

Stop. Look at the interval before you commit to an answer. If it has no ends, what settles whether a critical number is a Max or a min — and what is the name of that test? Then: which letter here is the variable, and which one is only riding along?

5. **The poster.** The top and bottom margins of a poster are each 6 cm and the side margins are each 4 cm. The area of printed material on the poster is fixed at 384 cm^2 . Find the dimensions of the poster with the smallest area.

Stop. Which rectangle is the 384 cm^2 one — the poster, or the printed part? Label both on your picture before you write a single slot.

At the boards

6. **BOARDS** **An open-top box.** A box with an open top is to be constructed from a square piece of cardboard, 3 ft wide, by cutting out a square from each of the four corners and bending up the sides. Find the largest volume that such a box can have.
- (a) Draw several diagrams illustrating the situation — some short boxes with large bases, some tall boxes with small bases. Find the volumes of several such boxes. Does it appear that there is a maximum volume? If so, estimate it.
 - (b) Draw a diagram illustrating the general situation. Introduce notation and label the diagram with your symbols.
 - (c) Write an expression for the volume.
 - (d) Use the given information to write an equation that relates the variables.
 - (e) Use part (d) to write the volume as a function of one variable.
 - (f) Finish solving the problem, and compare the answer with your estimate in part (a).

Put the template card on the wall beside your work, and put every part on the wall — including the guesses in part (a). They are what part (f) is compared against.